
Inferring Hidden Opponent Strategy Changes under Partial Observability in MOBA-like Environments

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Abstract

This project studies belief-based reasoning in partially observable environments with non-stationary opponents. Under such settings, the same observation sequence may be explained by multiple underlying strategies, making strategy inference fundamentally ambiguous. We propose a controlled MOBA-like environment in which opponent behavior is drawn from a predefined strategy library and may switch over time. The agent maintains a belief distribution over opponent strategies using Bayesian inference and uses this belief to support decision-making. To evaluate the role of belief-based reasoning, we compare belief-conditioned policies with observation-only baselines and introduce perturbed belief baselines to isolate the effect of correct inference. Our evaluation focuses on strategy inference accuracy, switch detection, and downstream decision performance. We further analyze under what conditions belief-based reasoning provides the greatest benefit. This work provides a controlled framework for understanding the role of reasoning in multi-agent environments with hidden and non-stationary dynamics.

1 Introduction

Partial observability and non-stationary opponents are two key challenges in multi-agent reinforcement learning (Hernandez-Leal et al. [2017, 2019], Papoudakis et al. [2019]). In many multi-agent environments, agents must make decisions based on incomplete information about the environment while interacting with other agents whose behavior may change over time. These challenges are especially prominent in competitive settings, where the policies of other agents may evolve or switch strategies during interaction.

A large body of research has studied decision-making under partial observability (Srinivasan et al. [2018]) and learning in non-stationary multi-agent environments separately. However, in many real-world strategic scenarios, agents must address both challenges simultaneously. In such settings, an agent must reason about hidden opponent behaviors while also adapting to potential changes in those behaviors over time.

Some recent approaches have begun to consider this joint setting by combining opponent modeling with partially observable environments (Zheng et al. [2018], Wang et al. [2022], Davies et al. [2020]). In many of these methods, opponent behaviors are assumed to be drawn from a bounded or predefined library of strategies. While this assumption simplifies the learning problem, it still leaves a fundamental challenge unresolved.

Under partial observability and stochastic policies, the same observation sequence may be explained by multiple strategies, making strategy inference fundamentally ambiguous. As a result, an agent must not only infer the opponent’s current strategy from interaction history, but also determine whether changes in observed behavior are due to stochastic variation or an actual shift in the underlying strategy.

This leads to a key research question: does explicitly modeling a belief over opponent strategies help resolve this ambiguity, enabling more accurate detection of strategy changes and improving decision-making under partial observability?

Understanding this is important for designing agents that can robustly operate in real-world strategic environments, where both uncertainty and non-stationarity are unavoidable.

We make the following contributions:

- We propose a controlled framework for studying belief-based reasoning under partial observability and non-stationary opponent dynamics.
- We isolate the effect of belief-based reasoning through a causal evaluation design, including perturbed belief baselines.
- We provide empirical analysis of when and under what conditions belief-based reasoning improves strategy inference, change detection, and downstream decision-making.

While prior work suggests that belief-based reasoning can improve decision-making under uncertainty, its practical impact on downstream performance remains unclear. In this work, we empirically evaluate whether belief inference improves policy learning in a partially observable, non-stationary environment.

Our results show that while belief models can infer opponent strategies and detect switching behavior, they do not necessarily improve task-level reward. Instead, belief signals provide modest improvements in behavioral stability and interpretability.

2 Problem Statement

We consider a partially observable multi-agent environment in which an agent interacts with an opponent whose underlying strategy is not directly observable and may change over time.

Let $\theta \in \Theta$ denote the opponent’s strategy, where Θ is a predefined strategy library. At each time step t , the agent receives an observation o_t and observes the opponent’s action, but does not have access to θ_t . The agent instead maintains a history of interactions $h_t = (o_1, a_1, \dots, o_t)$.

The goal of the agent is to infer the opponent’s current strategy from h_t and use this information to support decision-making. However, under partial observability and stochastic policies, multiple strategies may induce similar observation sequences. This makes the mapping from h_t to θ_t inherently ambiguous.

To address this, we consider maintaining a belief distribution over opponent strategies:

$$b_t = P(\theta \mid h_t)$$

which summarizes the agent’s inference over possible strategies given interaction history.

We study whether conditioning on this belief improves:

- strategy inference accuracy,
- detection of strategy changes,
- and downstream decision-making performance.

In addition, we explicitly evaluate whether the inferred belief improves downstream decision-making performance when integrated into a reinforcement learning policy.

3 Hypotheses

We hypothesize that explicitly modeling a belief over opponent strategies improves inference and decision-making in partially observable environments with non-stationary opponents.

H1 (Disambiguation): Belief-based methods achieve higher strategy inference accuracy than observation-only methods, particularly when multiple strategies induce similar observation sequences.

H2 (Change Detection): Belief-based agents detect opponent strategy switches with lower delay and greater stability than observation-based agents.

H3 (Environment Dependence): The benefit of belief modeling is most significant in environments with both partial observability and strategy switching, and diminishes when the opponent strategy is stationary.

4 Environment

We design a simplified MOBA-style multi-agent environment with two teams interacting in a partially observable setting. Each team consists of two agents (a laner and a jungler) operating in a symmetric map with a central lane and surrounding regions.

At each time step, agents receive partial observations o_t consisting of nearby units, visible opponents, and state variables (e.g., health, position, cooldowns). Observations are limited by fog of war.

Agents select actions from a discrete set, including movement, attacks, and ability usage. The environment evolves according to these joint actions.

Opponent behavior is governed by a latent strategy $\theta \in \Theta$, where each strategy corresponds to a stochastic policy. During an episode, the opponent may switch strategies at unknown time steps, inducing non-stationarity.

An episode terminates when a terminal condition is reached (e.g., objective completion or time limit).

5 Approach

Our framework consists of two main components: opponent modeling via belief inference, and belief-conditioned policy learning.

5.1 Opponent Strategy Model

We assume that opponent behavior is governed by a latent strategy $\theta \in \Theta$, where each strategy corresponds to a distinct behavioral pattern (e.g., aggressive, neutral, or farming).

Each strategy defines a stochastic policy over actions:

$$\pi_{\theta}(a \mid o) = \frac{\exp(\beta \cdot \text{score}_{\theta}(o, a))}{\sum_{a'} \exp(\beta \cdot \text{score}_{\theta}(o, a'))}$$

where o denotes the observation, a denotes an action, β controls the stochasticity, and $\text{score}_{\theta}(o, a)$ is a strategy-specific preference function.

In practice, the score function is implemented using heuristic features derived from the observation, such as relative position, target proximity, and combat context. These features encode prior assumptions about how different strategies behave in the environment.

5.2 Belief Update

The agent maintains a belief distribution over strategies:

$$b_t(\theta) = P(\theta \mid h_t)$$

where h_t denotes the interaction history up to time t .

We update the belief using Bayesian inference:

$$P(\theta \mid h_t) = \frac{P(h_t \mid \theta)P(\theta)}{\sum_{\theta'} P(h_t \mid \theta')P(\theta')}$$

The likelihood is approximated as:

$$P(h_t \mid \theta) \approx \prod_{k=1}^t \pi_{\theta}(a_k \mid o_k)$$

assuming conditional independence of actions given θ .

To improve computational stability, we perform updates in log-space and normalize the belief at each timestep. In addition, we use a smoothing factor to prevent the belief from collapsing too quickly to a single strategy.

5.3 Policy Learning

We train a policy conditioned on both observation and belief:

$$\pi(a_t \mid o_t, b_t)$$

The belief vector b_t is concatenated with the observation o_t and used as input to the policy network.

We optimize the policy using Proximal Policy Optimization (PPO) to maximize expected return:

$$J(\pi) = \mathbb{E} \left[\sum_{t=0}^T \gamma^t r_t \right]$$

We compare two policies:

- Observation-only policy $\pi(a_t \mid o_t)$
- Belief-conditioned policy $\pi(a_t \mid o_t, b_t)$

Importantly, the policy is not designed to perform opponent reasoning implicitly. Instead, it serves as a controlled setting to evaluate whether explicitly provided belief representations improve downstream decision-making.

5.4 Implementation Details

The policy is implemented as a two-head actor-critic multilayer perceptron using separate networks for the actor and critic. Each network consists of two hidden layers with 64 units and $\tanh$ activations. The actor outputs logits for a categorical distribution over 13 discrete actions, while the critic outputs a scalar value estimate.

For the observation-only baseline, the input is the environment observation vector. For the belief-conditioned policy, a belief vector over opponent strategies is concatenated with the observation, resulting in an increased input dimension.

We train the policy using PPO with a learning rate of 3×10^{-4} , discount factor $\gamma = 0.99$, and GAE parameter $\lambda = 0.95$. The PPO clipping parameter is set to 0.2, with value loss coefficient 0.5 and entropy coefficient 0.003. Gradients are clipped with a maximum norm of 0.5. Each update is performed over 4 epochs with 4 minibatches, using batches of 1024 environment steps.

Training is conducted for a total of 102,400 timesteps with a maximum episode length of 1000 steps. We use a single learning agent in a 1v1 setting with partial observability and a vision range of 5.0.

For expert-guided training, expert actions are injected with an initial probability of 0.9, which decays linearly to 0 over 80 updates. The expert policy follows an aggressive strategy, while the opponent uses a neutral strategy during baseline training.

For experiments involving non-stationarity, opponent strategies switch during the episode according to a time-based schedule, with switch points sampled uniformly between steps 150 and 850.

6 Evaluation

The proposed framework is evaluated along three dimensions: strategy inference, strategy switch detection, and downstream decision performance.

6.1 Strategy Inference

Let θ_t denote the ground-truth opponent strategy at time t , and let $b_t(\theta)$ denote the inferred belief distribution. The predicted strategy is defined as:

$$\hat{\theta}_t = \arg \max_{\theta} b_t(\theta)$$

Inference accuracy is measured as:

$$\text{Accuracy} = \frac{1}{T} \sum_{t=1}^T \mathbb{I}[\hat{\theta}_t = \theta_t]$$

Under higher levels of observation ambiguity, belief-based methods are expected to maintain significantly higher classification accuracy, whereas observation-only baselines are expected to degrade due to indistinguishable behavior patterns.

6.2 Strategy Switch Detection

Let t_{switch} denote the time step at which the opponent changes its strategy. The detection delay is defined as:

$$\Delta = \min\{t > t_{\text{switch}} : \arg \max_{\theta} b_t(\theta) = \theta_t\} - t_{\text{switch}}$$

Belief stability is evaluated during non-switching intervals by measuring fluctuations in b_t when θ_t remains constant.

We expect belief-based agents to detect strategy switches with lower delay and exhibit smoother belief transitions, whereas baseline methods may either respond more slowly or exhibit unstable fluctuations.

6.3 Decision Performance

Two policies are compared:

- Observation-only policy $\pi(a_t \mid o_t)$
- Belief-conditioned policy $\pi(a_t \mid o_t, b_t)$

Performance is evaluated using the following metrics:

- cumulative reward
- win rate

To isolate the effect of belief correctness, additional baselines with perturbed beliefs are introduced:

- Random belief
- Shuffled belief
- Lagged belief

We expect belief-conditioned policies to achieve higher cumulative reward and win rate, particularly in non-stationary environments. In contrast, when the opponent’s strategy is stationary, the performance gap between the two policies is expected to diminish.

These three evaluation dimensions allow us to separately assess (1) belief correctness and (2) its impact on downstream policy performance.

7 Results

7.1 Training Dynamics

We first analyze the training dynamics of PPO in the MOBA-like environment.

Figure 1a shows the evolution of action group distributions over training. Initially, the agent exhibits movement-dominated behavior, while attack actions are relatively infrequent. As training progresses, the agent gradually increases its attack frequency and reduces unnecessary movement, indicating that it learns to engage more effectively with the environment.

This behavioral shift is further supported by combat statistics in Figure 1b. The number of enemy minion takedowns increases significantly over training, while interactions with enemy agents remain sparse. This suggests that the agent learns basic lane-clearing behavior but does not consistently engage in higher-risk combat.

Reward-related metrics in Figure 1c show partial improvement but remain highly unstable. While the cumulative reward improves compared to early training stages, it exhibits high variance and remains predominantly negative. Advantage estimates also show large fluctuations, indicating noisy and inconsistent learning signals.

Despite this, optimization metrics in Figure 1d indicate stable training behavior. Both the PPO loss and value estimates decrease over time, and policy entropy gradually declines, suggesting that the policy is converging and exploration is being reduced.

However, these results highlight a key limitation: although PPO is able to learn structured behavior and achieve stable optimization, it fails to produce strong task-level performance in this environment.

To address this issue, we introduce expert-guided PPO, where expert actions are mixed into training with a decaying probability. As shown in later sections, expert guidance significantly improves learning stability and downstream performance. This suggests that exploration and credit assignment are major challenges in this environment, and that pure PPO is insufficient without additional guidance.

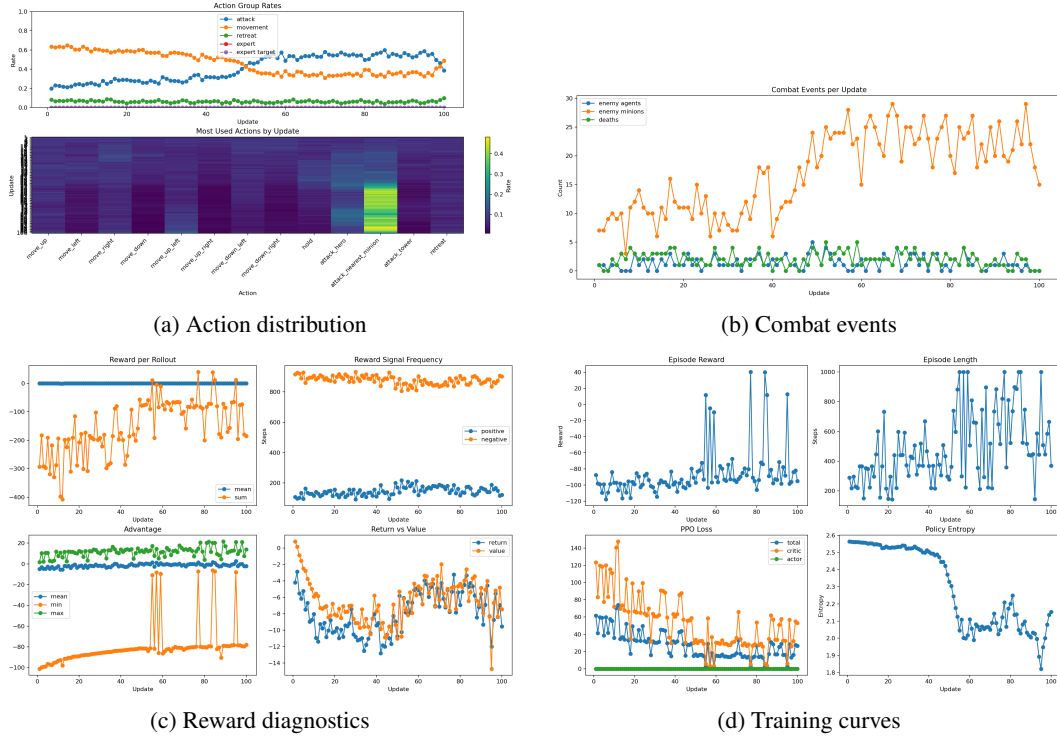


Figure 1: PPO training diagnostics. The agent gradually shifts toward more attack behavior and learns basic lane-clearing behavior, but reward remains unstable despite decreasing loss and entropy.

7.2 Baseline Selection

To evaluate the impact of belief-based reasoning, we first establish a strong observation-only baseline.

We compare multiple expert-guided PPO configurations, where expert actions are injected during training with a decaying probability. These configurations differ in the expert policy used for guidance (e.g., aggressive, neutral, farming) as well as the opponent setting during training.

Table 1 summarizes the performance of different configurations. Overall, expert-guided PPO significantly outperforms standard PPO, which fails to achieve positive reward across all opponent types. Among the expert-guided methods, configurations using aggressive expert guidance consistently achieve higher scores and final rewards.

Configuration	Score	Final Reward
aggressiveExpert_farming	343.2	117.9
aggressiveExpert_neutral	268.9	82.8
farmingExpert_neutral	237.5	67.8
neutralExpert_neutral	224.8	63.2
ppo_neutral	-65.8	-71.0

Table 1: Comparison of expert-guided PPO configurations.

Although aggressiveExpert_farming achieves the highest reward, it is specialized to a single opponent type. In contrast, aggressiveExpert_neutral provides a more balanced training signal and better generalization across different opponent behaviors. We therefore select aggressiveExpert_neutral as the baseline for subsequent experiments.

To better understand the effect of expert guidance, we visualize the training dynamics of the selected configuration.

As shown in Figure 2, expert-guided PPO achieves substantial improvement in episode reward over training. In addition, the action distribution gradually shifts toward more attack-oriented behavior, indicating that the agent learns structured and task-relevant policies under expert guidance.

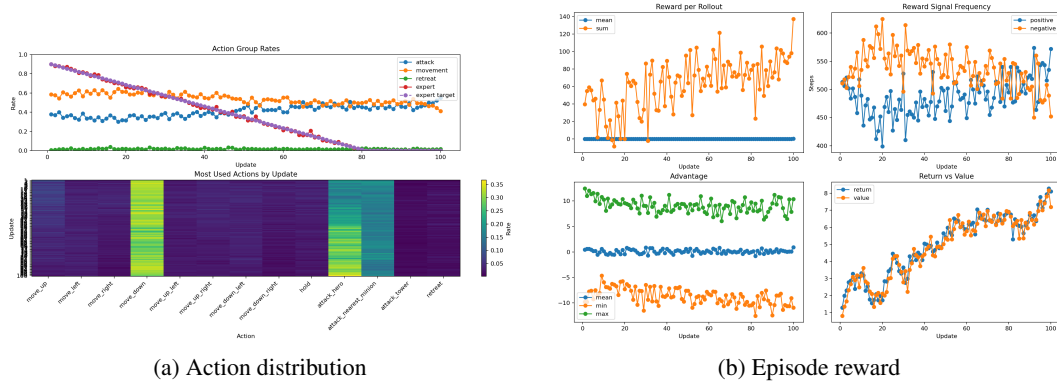


Figure 2: Training dynamics of expert-guided PPO. The agent achieves higher reward and develops more structured behavior compared to standard PPO.

7.3 Belief Correctness

Before evaluating whether belief improves policy performance, we first test whether the belief module can infer the opponent’s hidden strategy from observations.

At each timestep, we predict the opponent strategy using the maximum-probability belief:

$$\hat{\theta}_t = \arg \max_{\theta} b_t(\theta).$$

We then compare this prediction with the ground-truth scripted strategy.

Strategy	Accuracy
aggressive	0.8419
neutral	0.2645
farming	0.5096

Table 2: Belief inference accuracy for fixed opponent strategies.

As shown in Table 2, the belief model identifies aggressive behavior reliably, but performs worse on neutral and farming strategies. In particular, neutral is difficult to classify because its behavior overlaps with both aggressive and farming patterns under partial observability.

These results suggest that the belief module provides a meaningful but noisy opponent-strategy signal. Therefore, the following experiments should be interpreted as evaluating whether an imperfect belief representation can still improve downstream policy performance.

7.4 Downstream Policy Performance

We now evaluate whether incorporating belief improves downstream decision-making.

We compare two policies: an observation-only baseline trained with expert-guided PPO, and a belief-conditioned policy that additionally takes the belief distribution $b_t(\theta)$ as input.

Main Results Figure 3 compares observation-only and belief-conditioned policies.

Under fixed opponents, belief-conditioned policies do not achieve higher reward, and in some cases perform slightly worse. This suggests that simply providing belief as an additional input is insufficient for improving decision-making. Even when belief contains structured information about opponent strategy, the policy may fail to utilize this information effectively.

One possible explanation is that the observation-only policy already captures sufficient behavioral patterns through direct interaction, making the additional belief signal redundant. Alternatively, the policy network may not learn to associate variations in belief with meaningful changes in action, especially when the belief signal is noisy, as observed in Section 5.3.

However, under strategy-switching opponents, belief-conditioned policies exhibit improved stability. In particular, action distribution shifts are reduced and reward volatility decreases, indicating that belief provides a weak but consistent signal for adapting to non-stationary behavior. This suggests that belief may be more useful for stabilizing adaptation rather than improving raw performance.

Belief Ablation To further evaluate whether the policy utilizes belief, we conduct ablation experiments by replacing belief with randomized, shuffled, or lagged variants.

As shown in Figure 4, performance remains largely unchanged across these conditions. This indicates that the policy does not strongly depend on belief correctness. Even when the belief signal is degraded or randomized, the policy behavior remains largely unchanged.

This suggests that the policy has not learned to rely on belief as a primary decision-making signal, but instead continues to depend primarily on direct observations.

Summary Taken together, these results highlight a gap between inference and control: although the belief module captures structured information about opponent behavior, this information is not effectively exploited by the policy to improve reward.

This suggests that explicit belief modeling alone is insufficient. Future work should focus on tighter integration between belief inference and policy learning, such as end-to-end training or architectures that more directly couple belief dynamics with action selection.

8 Discussion

Our results provide several insights into the role of belief-based reasoning in partially observable, non-stationary environments.

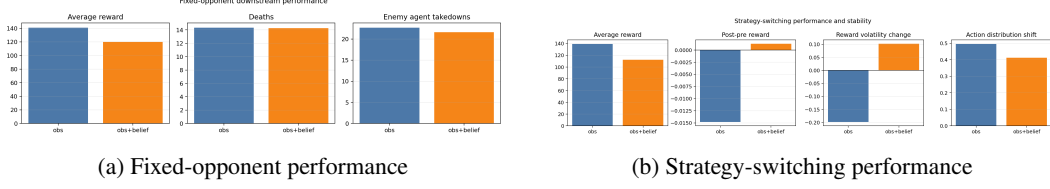


Figure 3: Downstream policy performance. Belief-conditioned policies do not improve reward under fixed opponents, but provide modest improvements in behavioral stability under strategy switching.

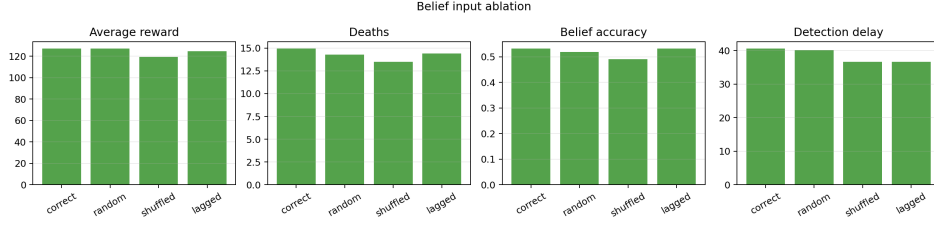


Figure 4: Belief input ablation. Replacing belief with random or shuffled signals does not significantly affect performance, indicating that the policy does not strongly depend on belief correctness.

Imperfect Belief as a Limitation First, the belief model itself is imperfect, particularly for neutral strategies. As shown in Section 5.3, neutral behavior is difficult to distinguish because it overlaps with both aggressive and farming patterns under partial observability. This ambiguity limits the quality of the belief signal and may reduce its usefulness for downstream decision-making.

Policy Does Not Utilize Belief More importantly, our ablation results suggest that the policy does not effectively utilize belief information. Even when the belief signal is randomized or degraded, policy performance remains largely unchanged. This indicates that the learned policy relies primarily on direct observations rather than explicitly inferred opponent strategies.

Observation May Be Sufficient One possible explanation is that the observation space already contains sufficient information for decision-making. Through interaction, the policy may implicitly capture behavioral patterns without requiring an explicit belief representation. In such cases, belief becomes redundant unless it provides information that is otherwise inaccessible.

Belief Helps Stability but Not Performance Despite not improving reward, belief-conditioned policies exhibit improved stability under strategy switching. This suggests that belief may serve as a regularizing signal that smooths policy responses to non-stationary behavior, rather than directly improving task performance.

Implications for RL and Multi-Agent Systems These findings highlight a gap between inference and control: even when structured information about the environment is available, it is not automatically exploited by the policy. This suggests that explicit reasoning modules must be more tightly integrated with policy learning in order to be effective.

Limitations This work has several limitations. The belief model relies on heuristic scoring functions rather than learned representations, which may restrict its expressiveness. In addition, the environment is simplified and focuses on a limited set of strategies. Finally, the policy architecture is relatively simple and may not be sufficient to fully leverage belief inputs.

Future Work Future work should explore end-to-end learning of belief representations, as well as architectures that more explicitly couple belief inference with policy updates. Extending the framework to more complex multi-agent settings, including cooperative scenarios and unseen strategies, would further test the usefulness of belief-based reasoning.

9 Conclusion

In this work, we studied belief-based opponent modeling in a partially observable, non-stationary environment.

We showed that belief inference can capture meaningful information about opponent strategies, particularly for clearly distinguishable behaviors such as aggressive play. However, belief accuracy remains limited in ambiguous cases such as neutral strategies.

More importantly, we found that incorporating belief into the policy does not improve downstream reward. Through ablation experiments, we observed that the policy does not strongly depend on belief correctness, suggesting that the inferred information is not effectively utilized.

Despite this, belief-conditioned policies exhibit improved stability under strategy switching, indicating that belief provides a weak but consistent signal for handling non-stationarity.

Overall, our results suggest that while belief modeling provides interpretable structure, it is not sufficient on its own to improve decision-making. Bridging the gap between inference and control remains an important direction for future research in reinforcement learning and multi-agent systems.

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